

The Role of Generative AI in the AWS Database Migration Service

The integration of generative AI in the AWS Database Migration Service enhances schema conversion, making data migrations more accurate, speedy, and efficient.



In today's rapidly shifting digital landscape, organisations are compelled to modernise their data infrastructure to stay competitive and agile. Database migration is an essential part of this transformation, yet it presents a range of challenges—complex schemas, diverse technologies, and the risk of downtime. Amazon Web Services (AWS) offers the Database Migration Service (DMS) to streamline this process, and recent advancements in generative AI are further revolutionising how migrations are carried out.

Homogeneous vs heterogeneous database migration

Homogeneous database migration refers to the movement of data between two databases that share the same engine or technology. For instance, migrating from MySQL to MySQL, or Oracle to Oracle. These migrations typically involve fewer compatibility issues because the source and target databases speak the same 'language', making schema conversion and data transfer relatively straightforward.

Consider the scenario of an organisation running its production workload on Amazon RDS MySQL. To improve performance and scalability, the business decides to migrate to Amazon Aurora MySQL, a high-availability, cloud-native database service.

AWS DMS facilitates this process by connecting to the source RDS MySQL instance, extracting and transforming the data as needed, and loading it into Aurora MySQL. Since both databases share the same engine, schema conversion is usually direct, and the risk of data loss or incompatibility is minimal.

Heterogeneous database migration involves moving data between databases that use different engines or technologies. Examples include migrating from Oracle to PostgreSQL, or SQL Server to MySQL. These migrations are inherently more complex, as differences in data types, functions, and procedural code must be reconciled to ensure seamless compatibility.

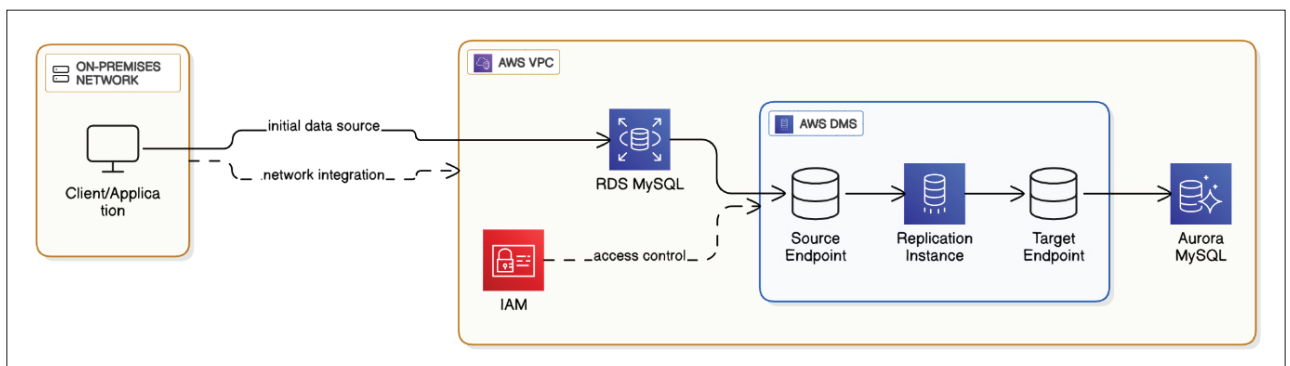


Figure 1: AWS reference architecture for homogeneous data migration